

CHM111 Practice Exam on chapter(s): 3, 4

1. 1.90×10^1 g
2. Oxidized: Rb Reduced: Cd
3. 9.75×10^{-7} g
4. AB_4
5. $\text{Ca}(s) + 2 \text{H}_2\text{O}(l) \rightarrow \text{Ca}(\text{OH})_2(aq) + \text{H}_2(g)$
6. 88.2%
7. 21.55 g
8. $2 \text{ZnS} + 2 \text{C} + 3 \text{O}_2 \rightarrow 2 \text{CO} + 2 \text{SO}_2 + 2 \text{Zn}$
9. Nitrogen
10. 18.1%
11. 2.04×10^{10} g
12. $\text{HBr}(aq) + \text{H}_2\text{O}(\ell) \rightarrow \text{Br}^-(aq) + \text{H}_3\text{O}^+(aq)$
13. 1.44 moles
14. 98.0 g/mol
15. 85.40 mL
16. ionic
17. $\text{Ag}^+(aq) + \text{F}^-(aq) \rightarrow \text{AgF}(s)$
18. 1.21 moles
19. 141.9 mL
20. 4.29×10^1 g
21. $\text{C}_8\text{H}_{17}\text{OH}$
22. $3 \text{H}_2 + 4 \text{O}_2 + \text{N}_2 \rightarrow 2 \text{HNO}_3 + 2 \text{H}_2\text{O}$
23. 64%
24. molecular
25. 9
26. 6.859 g
27. 2.36×10^{24} germanium atoms
28. 42%
29. 61.04 g
30. reduced
31. $6 \text{MnCO}_3 + 8 \text{Al} + 3 \text{O}_2 \rightarrow 4 \text{Al}_2\text{O}_3 + 6 \text{CO}_2 + 6 \text{Mn}$
32. 0.010891 moles
33. Single-Displacement
34. 83.2 mM
35. 3.808×10^{26} g
36. 12.29 g
37. 0.9894 mL of lithium phosphate
38. $2 \text{H}_3\text{AsO}_4 + 3 \text{Co}(\text{OH})_2 \rightarrow 6 \text{H}_2\text{O} + \text{Co}_3(\text{AsO}_4)_2$
39. $109.9 \frac{\text{g}}{\text{mol}}$
40. $0.831 \frac{\text{mol}}{\text{L}}$
41. 0.0264 mol
42. 9
43. $\text{TiO}_2 + 2 \text{Cl}_2 + \text{C} + 2 \text{Mg} \rightarrow \text{Ti} + \text{CO}_2 + 2 \text{MgCl}_2$
44. $3 \text{C} + 4 \text{Al}(\text{OH})_3 \rightarrow 4 \text{Al} + 6 \text{H}_2\text{O} + 3 \text{CO}_2$
45. AlPO_4 and H_2O
46. BN
47. No reaction occurs.
48. $\text{Fe}_2\text{O}_3 + 5 \text{CO} \rightarrow 2 \text{Fe} + 5 \text{CO}_2$
49. 65.2%
50. 89. g